

HW3 Solutions

1. Problem 1-3

Simplify the following expressions using Boolean algebra.

- a. $A + AB$
- b. $AB + AB'$
- c. $A'BC + AC$
- d. $A'B + ABC' + ABC$

Solution:

- a. $A + AB = A(1+B) = A$
- b. $AB + AB' = A(B+B') = A$
- c. $A'BC + AC = C(A'B+A) = C(A'+A)(B+A) = C(A+B)$
- d. $A'B + ABC' + ABC = A'B + AB(C+C') = A'B + AB = B(A'+A) = B$

2. Problem 1-5

Using DeMorgan's theorem, show that:

- a. $(A + B)'(A' + B')' = 0$
- b. $A + A'B + A'B' = 1$

Solution:

- a. $(A + B)'(A' + B')' = (A'B')(AB) = 0$
- b. $A + A'B + A'B' = A + A'(B + B') = A + A' = 1$

3. Problem 1-7

Given the Boolean function

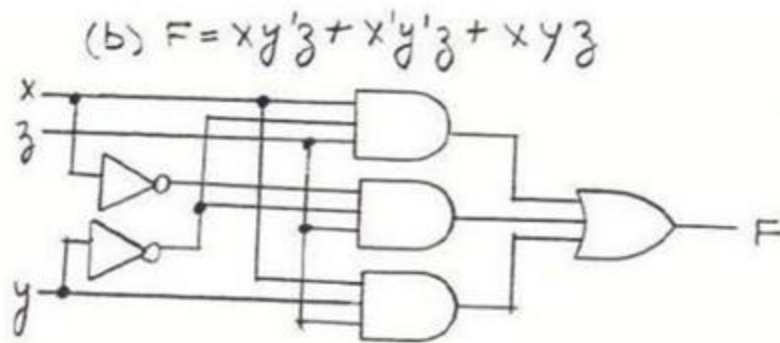
$$F = xy'z + x'y'z + xyz$$

- a. List the truth table of the function.
- b. Draw the logic diagram using the original Boolean expression.
- c. Simplify the algebraic expression using Boolean algebra.
- d. List the truth table of the function from the simplified expression and show that it is the same as the truth table in part (a).
- e. Draw the logic diagram from the simplified expression and compare the total number of gates with the diagram of part (b).

Solution:

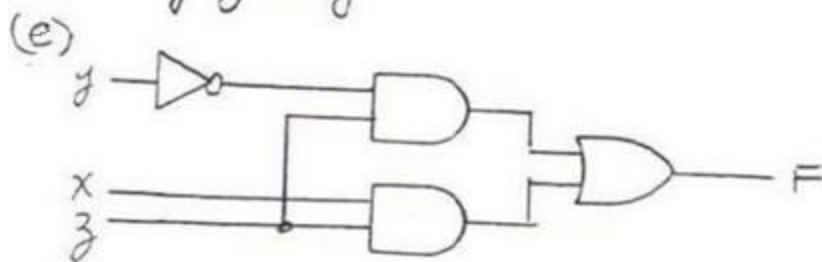
(a)

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1



(c) $F = xy'z + x'y'z + xyz$
 $= y'z(x + x') + xz(y + y')$
 $= y'z + xz$

(d) Same as (a)



4. Simplify the following Boolean functions:

- $F = AB + BC + B'C$
- $F = A + A'B$
- $F = A'B'C + A'BC + AB'$
- $F = AB + (AC)' + AB'C(AB + C)$
- $F = ((XY' + XYZ)' + X(Y + XY'))'$

Solution:

4a. $F = AB + BC + B'C = AB + C(B + B') = AB + C$

4b. $F = A + A'B = (A + A')(A + B) = A + B$

$$4c. \quad F = A'B'C + A'BC + AB' = A'C(B' + B) + AB' = A'C + AB'$$

$$\begin{aligned}
 4d. \quad F &= AB + (AC)' + AB'C(AB + C) \\
 &= AB + A' + C' + AB'C \cdot AB + AB'C \cdot C \\
 &= AB + A' + C' + 0 + AB'C(B \cdot B' = 0 \text{ and } C \cdot C = C) \\
 &= ABC + ABC' + A' + C' + AB'C(AB = AB(C + C') = ABC + ABC') \\
 &= AC(B + B') + C'(AB + 1) + A' \\
 &= AC + C' + A'(B + B' = 1 \text{ and } AB + 1 = 1) \\
 &= AC + (AC)' \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 4e. \quad F &= ((XY' + XYZ)' + X(Y + XY'))' \\
 &= ((X(Y' + YZ))' + XY + XY')' \\
 &= ((X(Y'Z + Y' + YZ))' + X(Y + Y'))' \quad (Y' = Y'(Z + 1) = Y'Z + Y') \\
 &= (X(Y' + Z))' + X' \\
 &= (X' + (Y' + Z)' + X)' \\
 &= (1 + YZ')' \\
 &= 1' \\
 &= 0
 \end{aligned}$$

5. Obtain the sum of product form of the following functions using Minterms.

a. $F(A, B, C) = A + BC$

b. $F(A, B, C, D) = AB + ACD$

Solution:

$$\begin{aligned}
 5a. \quad F(A, B, C) &= A + BC \\
 &= A(B + B')(C + C') + BC(A + A') \\
 &= (AB + AB')(C + C') + ABC + A'BC \\
 &= ABC + AB'C + ABC' + AB'C' + ABC + A'BC \\
 &= ABC + AB'C + ABC' + AB'C' + A'BC \quad (\text{as } ABC + ABC = ABC)
 \end{aligned}$$

5b.

$$\begin{aligned}
 F(A, B, C, D) &= AB + ACD \\
 &= AB(C + C')(D + D') + ACD(B + B') \\
 &= (ABC + ABC')(D + D') + ABCD + AB'CD \\
 &= ABCD + ABCD' + ABC'D + ABC'D' + ABCD + AB'CD \\
 &= ABCD + ABCD' + ABC'D + ABC'D' + AB'CD
 \end{aligned}$$

6. Obtain the product of sums form of the following functions using Maxterms.
 a. $F(A, B, C) = A + B'C$

Solution:

$$\begin{aligned}
 F(A, B, C) &= A + B'C \\
 &= (A + B')(A + C) && \text{[using distributive law]} \\
 &= (A + B' + CC')(A + C + BB') \\
 &= (A + B' + C)(A + B' + C')(A + B + C)(A + B' + C) \\
 &&& \text{[using the distributive property, as } X + YZ = (X + Y)(X + Z)\text{]} \\
 \text{Now, } (A + B' + C)(A + B' + C) &= A + B' + C; \\
 F &= (A + B' + C)(A + B' + C')(A + B + C)
 \end{aligned}$$

7. Find the complement of each of the functions by applying DeMorgan's theorem as many times as necessary

- a. $F1 = X'YZ' + X'Y'Z$
 b. $F2 = X(Y'Z' + YZ)$

Solution:

$$\begin{aligned}
 7a. \quad F1 &= X'YZ' + X'Y'Z = X'(YZ' + Y'Z) \\
 F1' &= X + (Y \oplus Z)' = X + Y \odot Z
 \end{aligned}$$

$$F2 = X(Y'Z' + YZ)$$

$$7b. \quad F2' = X' + (Y'Z' + YZ)' = X' + (Y \odot Z)' = X' + Y \oplus Z$$

8. Simplify the following expression using Boolean algebra:
 a. $A'BC + AB'C + ABC' + ABC$
 b. $(A' + B)' + B(A' + AC) + ABC'$

Solution:

$$\begin{aligned}
 8a. \quad &A'BC + AB'C + ABC' + ABC \\
 &= A'BC + AB'C + ABC' + ABC + ABC + ABC \\
 &= A'BC + ABC + AB'C + ABC + ABC' + ABC \\
 &= BC(A' + A) + AC(B' + B) + AB(C' + C) \\
 &= BC + AC + AB
 \end{aligned}$$

8b.

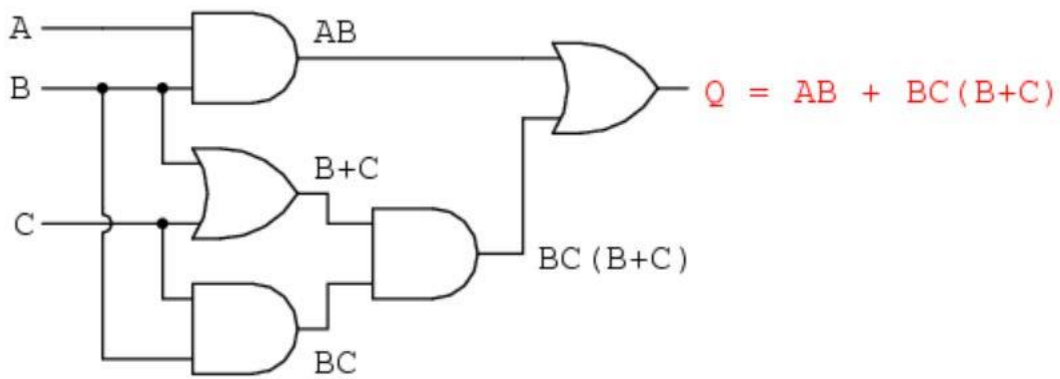
$$\begin{aligned}
 & (A'+B)'+B(A'+AC)+ABC' \\
 &= AB'+BA'+BAC+ABC' \\
 &= AB'+BA'+AB(C+C') \\
 &= AB'+BA'+AB \\
 &= AB'+B(A+A') \\
 &= AB'+B \\
 &= A+B
 \end{aligned}$$

9. Construct a logic diagram for the given Boolean expression:

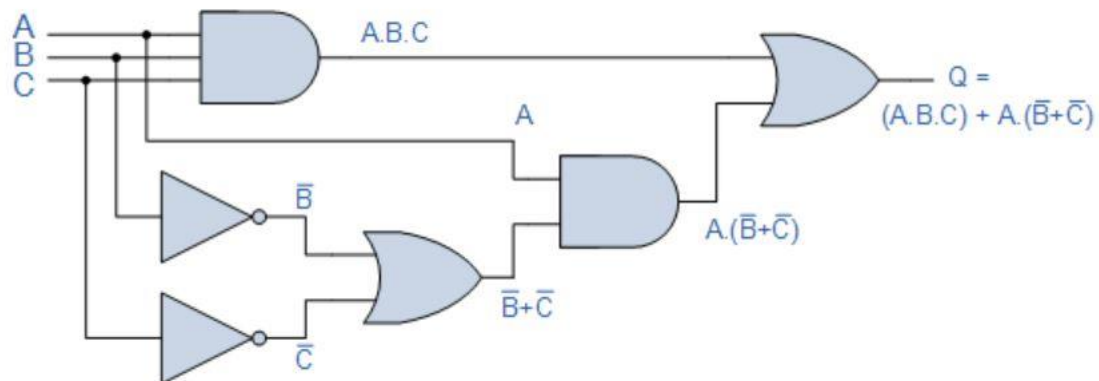
- $AB + BC(B + C)$
- $ABC + A(B' + C')$

Solution:

9a)



9b)



10. Find the dual of the function $F = xyz + x'yz' + y'z$

Solution:

$$(x + y + z)(x' + y + z')(y' + z)$$

11. Find the dual of the function $F = AB (C + (DL' G (B' + A + E))) (H + (J'A'B))$

Solution:

$$A + B + (C (D + L' + G + (B'AE))) + (H (J' + A' + B))$$

12. Implement the function (Draw the circuit diagram) of the following truth table using Sum-of-Products form

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Solution:

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

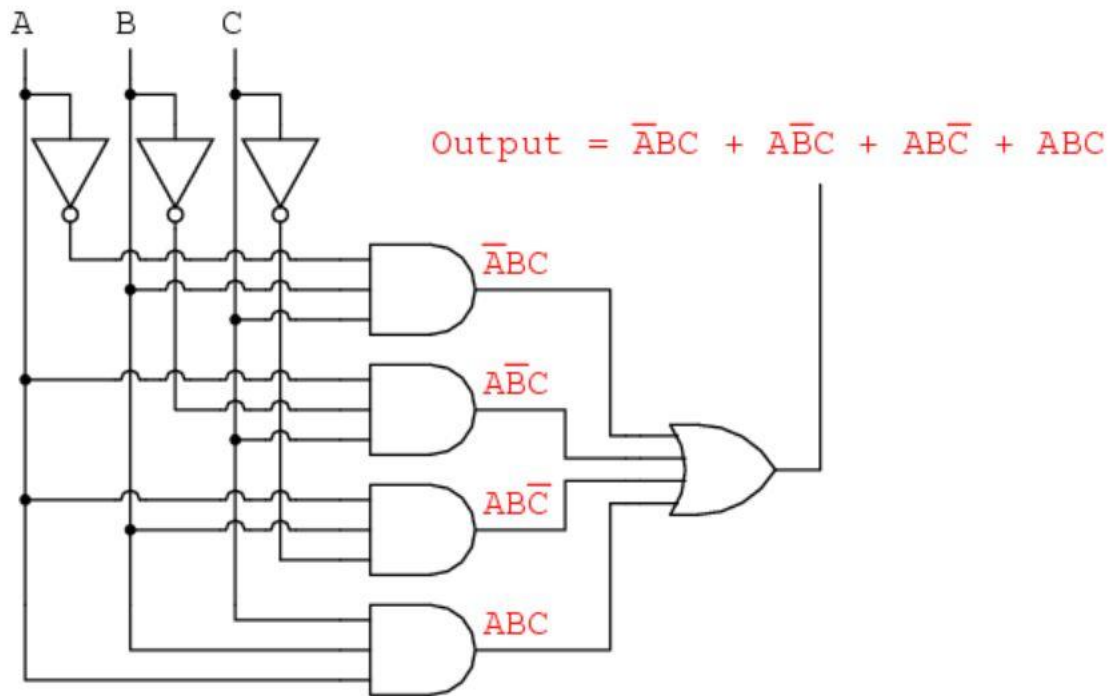
$$\overline{A}BC = 1$$

$$A\overline{B}C = 1$$

$$AB\overline{C} = 1$$

$$ABC = 1$$

$$\text{Output} = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$$



13. Implement the function (Draw the circuit diagram) of the following truth table using Sum-of-Products form:

a	b	c	d	z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Solution:

At first, find the all the rows that 'z' output equal to '1'. Then get the sum of products form of these rows. Now we get the Z equation:

$$\begin{aligned}
 z &= a'b'cd \\
 &+ a'bcd \\
 &+ ab'cd \\
 &+ abc'd' \\
 &+ abc'd \\
 &+ abcd' \\
 &+ abcd \\
 &= ab + cd
 \end{aligned}$$

Detailed simplification:

$$Z = ab(c'd' + c'd + cd' + cd) + cd(a'b' + a'b + ab')$$

$$= ab(c'd' + c'd + c) + cd(a'b' + a'b + ab')$$

$$= ab(c'd' + c + d) + cd(a'b' + a'b + ab')$$

Using DeMorgan's Law

$$= ab(c' + d' + c + d) + cd(a'b' + a'b + ab')$$

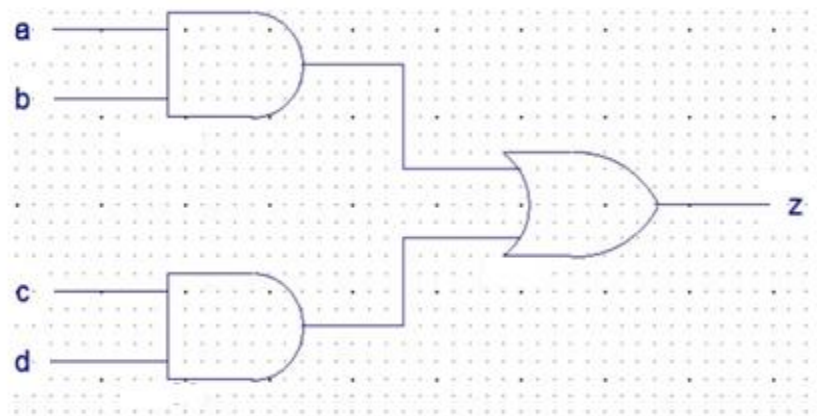
$$= ab + cd(b' + a'b)$$

$$= ab + cd(b' + a')$$

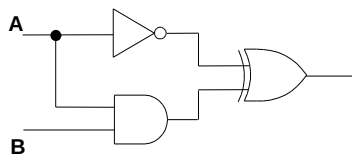
$$= ab + cd$$

The logic circuit is shown below:

$$z = ab + cd$$



14. Show the behavior of the following circuit using a truth table:

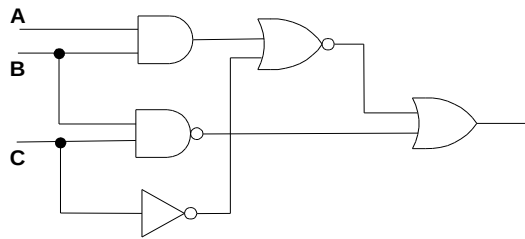


Solution:

The true table from the circuit is as follows,

A	B	A'	AB	$A' \oplus (AB)$
0	0	1	0	1
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1

15. Show the behavior of the following circuit using a truth table:



Solution:

The true table from the circuit is as follows,

A	B	C	AB	$(BC)'$	C'	$(AB+C')'$	$(BC)' + (AB+C')' = B' + C' + (A' + B')C$
0	0	0	0	1	1	0	1
0	0	1	0	1	0	1	1
0	1	0	0	1	1	0	1
0	1	1	0	0	0	1	1
1	0	0	0	1	1	0	1
1	0	1	0	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	0	0	0	0